

2.6 Linear Combinations of Harmonically Related Complex Exponentials

Previously, it was said that a signal $x(t)$ is periodic if there exists some positive value T such that:

$$x(t) = x(t + T) \text{ for all } t. \quad (2.6)$$

The smallest number of T which satisfies (2.6) is by definition, the fundamental period, and the fundamental frequency is defined as $\omega_0 = 2\pi / T$.

2 basic periodic signals were introduced last week, the sinusoidal signal

$$x(t) = \cos(\omega_0 t) \quad (2.7)$$

and the periodic complex exponential

$$x(t) = e^{j\omega_0 t}. \quad (2.8)$$

Both signals have a **fundamental frequency**, and **fundamental period** of ω_0 , and $T = 2\pi / \omega_0$ respectively.

Associated with the signal in eq. (2.8) is the set of **harmonically related** complex exponentials

$$\phi_k(t) = e^{jk\omega_0 t} = e^{jk(2\pi/T)t}, \quad k = 0, \pm 1, \pm 2, \dots \quad (2.9)$$

Each of these signals has a fundamental frequency that is a multiple of ω_0 , and therefore, each is periodic with period T (although for $|k| \geq 2$, the fundamental period of $\phi_k(t)$ is a fraction of T). Thus, a linear combination of harmonically related complex exponentials of the form

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \quad (2.10)$$

is also periodic with period T . In eq.(2.10), the term for $k = 0$ is constant. The terms when $k = +1$, $k = -1$ and are called the **fundamental components** or the **first harmonic components**, and have periods of T (or equivalently, fundamental frequencies of ω_0). The terms when $k = +2$ and $k = -2$, are called the **second harmonic components**, and have fundamental periods of $T/2$ (or equivalently, a frequencies of $2\omega_0$). Generally speaking, the components corresponding to $k = +N$ and $k = -N$ are called the **N^{th} harmonic components**.

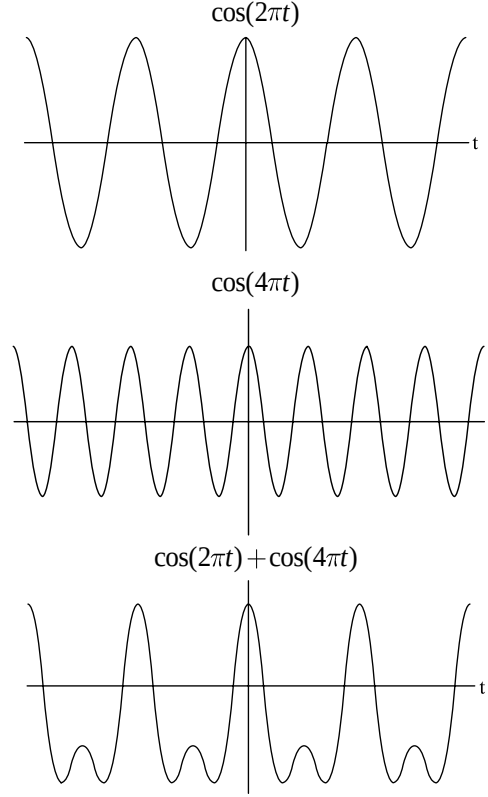


Figure 2.16. $\cos(2\pi t)$, $\cos(4\pi t)$ and its sum $\cos(2\pi t) + \cos(4\pi t)$.

The representation of a periodic signal in the form of eq.(2.10) is referred to as the **Fourier series** representation. Before developing the properties of this representation, let us consider an example.

Example 2.7

Consider a periodic signal $x(t)$, with fundamental frequency 2π , that is expressed in the form of eq.(2.10), as

$$x(t) = \sum_{k=-3}^{+3} a_k e^{jk2\pi t}, \quad (2.11)$$

where

$$\begin{aligned} a_0 &= 2, & a_2 &= a_{-2} = \frac{1}{3}, \\ a_1 &= a_{-1} = \frac{1}{2}, & a_3 &= a_{-3} = \frac{1}{4}. \end{aligned}$$

Rewriting eq.(2.11) and collecting each of the harmonic components which have the same fundamental frequency, we obtain

$$x(t) = 2 + \frac{1}{2}(e^{j2\pi t} + e^{-j2\pi t}) + \frac{1}{3}(e^{j4\pi t} + e^{-j4\pi t}) + \frac{1}{4}(e^{j6\pi t} + e^{-j6\pi t}). \quad (2.12)$$

Equivalently, using Euler's relation, we can write $x(t)$ in the form

$$x(t) = 2 + \cos 2\pi t + \frac{2}{3} \cos 4\pi t + \frac{1}{2} \cos 6\pi t. \quad (2.13)$$

In Figure 2.17, we illustrate graphically how the signal $x(t)$ is built up from its harmonic components.

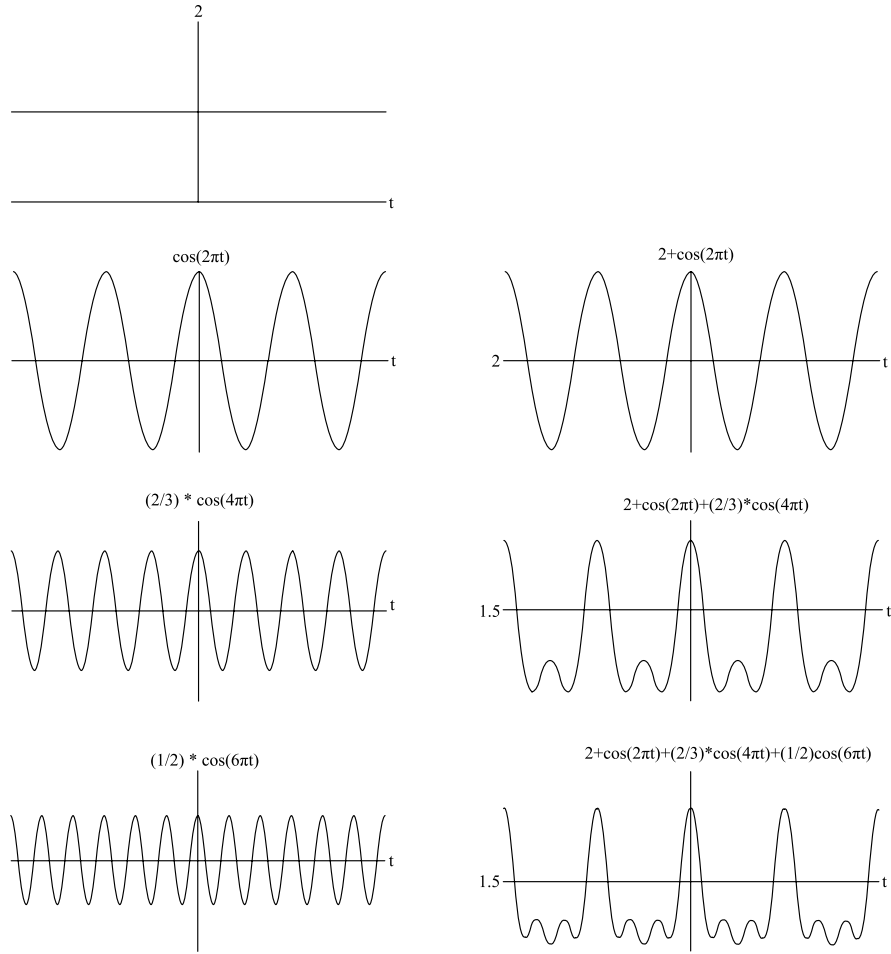


Figure 2.17. Construction of the signal $x(t)$ in Example 2.7 as a linear combination of harmonically related sinusoidal signals.

Example 2.8

Let

$$x(t) = 1 + \sin \omega_0 t + 2 \cos \omega_0 t + \cos \left(2\omega_0 t + \frac{\pi}{4} \right).$$

which has fundamental frequency ω_0 . As with the previous example, we can again expand $x(t)$ directly in terms of complex exponentials, so that

$$x(t) = 1 + \frac{1}{2j} [e^{j\omega_0 t} - e^{-j\omega_0 t}] + [e^{j\omega_0 t} + e^{-j\omega_0 t}] + \frac{1}{2} [e^{j(2\omega_0 t + \pi/4)} + e^{-j(2\omega_0 t + \pi/4)}].$$

Collecting terms, we obtain

$$x(t) = 1 + \left(1 + \frac{1}{2j} \right) e^{j\omega_0 t} + \left(1 - \frac{1}{2j} \right) e^{-j\omega_0 t} + \left(\frac{1}{2} e^{j(\pi/4)} \right) e^{j2\omega_0 t} + \left(\frac{1}{2} e^{-j(\pi/4)} \right) e^{-j2\omega_0 t}.$$

Thus, the Fourier series coefficients for this example are

$$\begin{aligned} a_0 &= 1, \\ a_1 &= \left(1 + \frac{1}{2j} \right) = 1 - \frac{1}{2}j, \\ a_{-1} &= \left(1 - \frac{1}{2j} \right) = 1 + \frac{1}{2}j, \\ a_2 &= \frac{1}{2} e^{j(\pi/4)} = \frac{\sqrt{2}}{4} (1 + j), \\ a_{-2} &= \frac{1}{2} e^{-j(\pi/4)} = \frac{\sqrt{2}}{4} (1 - j), \\ a_k &= 0, \quad |k| > 2. \end{aligned}$$

In Figure 2.18, we show a bar graph of the magnitude and phase of a_k as a function of k . Alternatively, one can plot the a_k as a function of frequency.

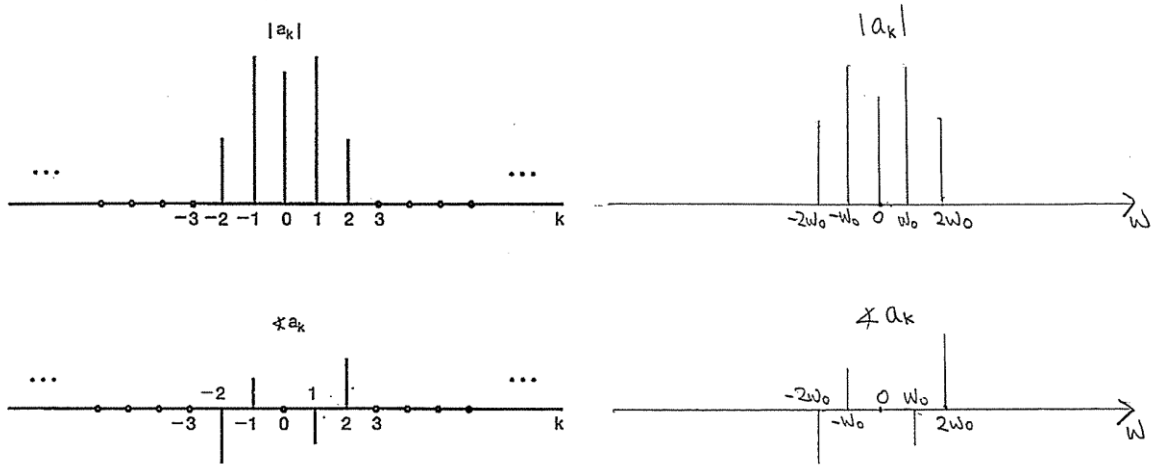


Figure 2.18 Plots of the magnitude and phase of the Fourier coefficients of the signal considered in **Example 2.8**.

We are leading towards something much more general. Actually, any periodic function $x(t)$ of period T can be expressed as an infinite sum of complex exponentials. This summation is called **Fourier series**. Fourier series can be written in several forms. One such form is the **exponential Fourier series**:

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$

where

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt \quad k = 0, \pm 1, \pm 2, \dots \quad (2.14)$$

Proof: If

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t},$$

we can multiply both sides by $e^{-jk\omega_0 t}$ to obtain

$$x(t) e^{-jk\omega_0 t} = \sum_{n=-\infty}^{\infty} a_n e^{j\omega_0(n-k)t} = \sum_{\substack{n=-\infty \\ n \neq k}}^{\infty} a_n e^{j\omega_0(n-k)t} + a_k.$$

Integrate both sides from $t = -T/2$ to $t = T/2$,

$$\begin{aligned} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt &= \int_{-T/2}^{T/2} \sum_{\substack{n=-\infty \\ n \neq k}}^{\infty} a_n e^{j\omega_0(n-k)t} dt + \int_{-T/2}^{T/2} a_k dt \\ &= \sum_{\substack{n=-\infty \\ n \neq k}}^{\infty} a_n \frac{1}{j\omega_0(n-k)} e^{j\omega_0(n-k)t} \Bigg|_{-T/2}^{T/2} + Ta_k \\ &= \sum_{\substack{n=-\infty \\ n \neq k}}^{\infty} a_n \frac{1}{j\omega_0(n-k)} [e^{j\pi(n-k)} - e^{-j\pi(n-k)}] + Ta_k \\ &= \sum_{\substack{n=-\infty \\ n \neq k}}^{\infty} a_n \frac{1}{j\omega_0(n-k)} [-1 - (-1)] + Ta_k \\ &= Ta_k \end{aligned}$$

If $x(t)$ has period T , the term

$$a_1 e^{j\omega_0 t} = a_1 e^{j2\pi t/T}$$

is referred to as the fundamental frequency and the other terms are known as the harmonics of $x(t)$. Note that the integration limits in calculating the Fourier coefficients a_k doesn't have to be from $-T/2$ to $T/2$: it can be from 0 to T , T to $2T$, $-3.5T$ to $-2.5T$ etc as long as it covers one period of $x(t)$

Perhaps one of the major applications of the concept of Fourier Series is on sound analysis. In fact, for a given pitch (or frequency), different instruments sound differently and we are able to distinguish whether it is a tone played by a piano, guitar, violin, saxophone. Why is that? The following are sample waveforms of a sound with the same pitch played by different instruments.

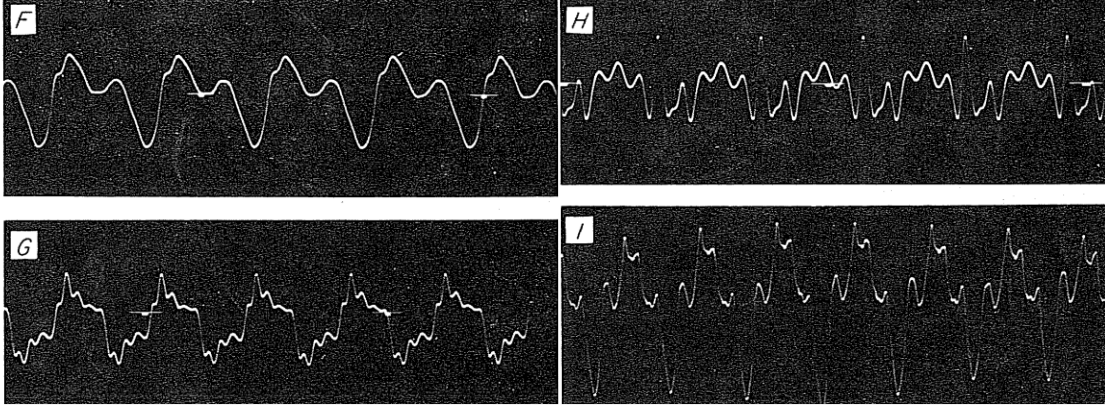


Figure 2.19 Waveform of a note played by F) a flute; G) a clarinet; H) an oboe and I) a saxophone.

Graphically, we can tell that even though the pitch (frequency) of the note is the same, the waveform of sound played by different instruments look differently. Therefore, one would expect that our ear can distinguish which instrument the note is played by based on the waveform differences. Mathematically, for a periodic waveform

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn\omega_0 t}$$

with the same fundamental pitch (frequency) ω_0 , the values of the Fourier Series coefficients $(\dots a_{-2}, a_{-1}, a_0, a_1, a_2, \dots)$ are generally different for each instrument. Therefore, the values of the Fourier Series coefficients $(\dots a_{-2}, a_{-1}, a_0, a_1, a_2, \dots)$ relative to each other is what distinguish one instrument from another. In music, such a feature is called the timbre, or quality of a musical tone.

Example 2.9

The periodic square wave, sketched in Figure 2.20 and defined over one period as

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases} \quad (2.15)$$

is a signal that we will encounter a number of times throughout this book. This signal is periodic with fundamental period T and fundamental frequency $\omega_0 = 2\pi/T$.

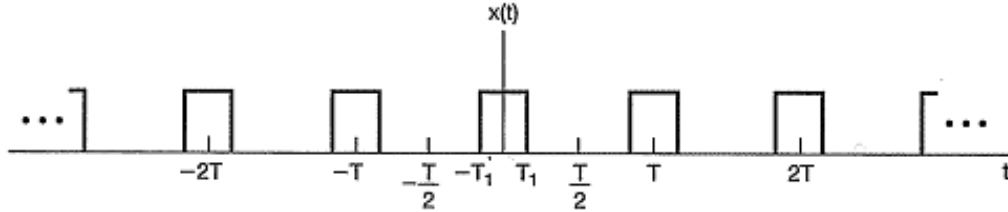


Figure 2.20 Periodic square wave with period T .

To determine the Fourier series coefficients for $x(t)$, we can use (2.14). Because of the symmetry of $x(t)$ about $t = 0$, it is convenient to choose $-T/2 \leq t \leq T/2$ as the interval over which the integration is performed, although any

interval of length T is equally valid and thus will lead to the same result. Using these limits of integration and substituting from eq.(2.15), we have first, for $k = 0$,

$$a_0 = \frac{1}{T} \int_{-T_1}^{T_1} dt = \frac{2T_1}{T}.$$

As mentioned previously, a_0 is interpreted to be the average value of $x(t)$, which in this case equals the fraction of each period during which $x(t) = 1$. For $k \neq 0$, we obtain

$$a_k = \frac{1}{T} \int_{-T_1}^{T_1} e^{-jk\omega_0 t} dt = -\frac{1}{jk\omega_0 T} e^{-jk\omega_0 t} \Big|_{-T_1}^{T_1},$$

which we may re-write as

$$a_k = \frac{2}{k\omega_0 T} \left[\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{2j} \right].$$

Noting that the term in brackets is $\sin k\omega_0 T_1$, we can express the coefficients a_k as

$$a_k = \frac{2 \sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(2k\pi T_1 / T)}{k\pi} = \frac{2T_1}{T} \frac{\sin(2k\pi T_1 / T)}{2k\pi T_1 / T} = \frac{2T_1}{T} \text{sinc}(2kT_1 / T), \quad k \neq 0, \quad (2.16)$$

where we have used the fact that $\omega_0 T = 2\pi$.

Figure 2.21 is a bar graph of the Fourier series coefficients for this example. In particular, the coefficients are plotted for a fixed value of T_1 and several values of T . For this specific example, the Fourier coefficients are real, and consequently, they can be depicted graphically with only a single graph. More generally, of course, the Fourier coefficients a_k are complex, so the two graphs corresponding to magnitude $|a_k|$ and phase $\angle a_k$ would be required. The plot $|a_k|$ is called **spectral amplitude** and represents the amplitude of n^{th} harmonic and the plot $\angle a_k$ is called the **spectral phase**. Graphic representation of $|a_k|$ along with $\angle a_k$ is called **complex frequency spectrum**. If a_k is purely real or purely imaginary, we can disregard the phase spectrum. The exponential Fourier series find extensive applications in communication theory. It is equivalent to resolving the function in terms of harmonically related frequency components of a fundamental frequency.

Lets set $T = 4T_1$ i.e. $x(t)$ is a square wave that is unity for half the period and zero for half the period. In this case, $\omega_0 T_1 = \pi / 2$, $T_1 / T = 1 / 4$ and from eq.(2.16),

$$a_k = \frac{1}{2} \text{sinc}\left(\frac{k}{2}\right) = \frac{1}{2} \frac{\sin(\pi k / 2)}{\pi k / 2} = \frac{\sin(\pi k / 2)}{\pi k}, \quad k \neq 0, \quad (2.17)$$

while

$$a_0 = \frac{1}{2}. \quad (2.18)$$

From eq. (2.17), $a_k = 0$ for k even and nonzero. Also $\sin(\pi k / 2)$ alternates between +1 and -1 for successive odd values of k . Therefore,

$$a_1 = a_{-1} = \frac{1}{\pi}, \quad a_3 = a_{-3} = -\frac{1}{3\pi}, \quad a_5 = a_{-5} = \frac{1}{5\pi} \quad \dots$$

The plot of Fourier Series coefficients a_k as a function of k or ω is known as the **spectrum** of the signal $x(t)$.

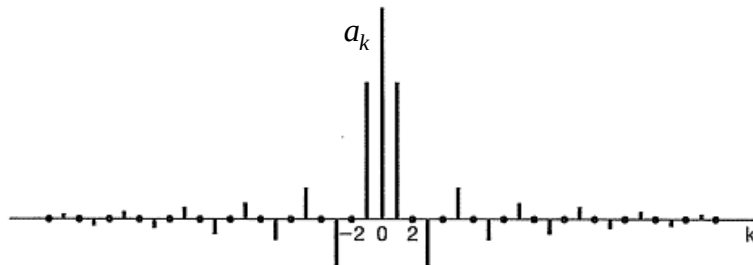


Figure 2.21. Plots of the Fourier series coefficients a_k for the periodic square wave with $T = 4T_1$. The coefficients are regularly spaced samples of the envelope $(2 \sin \omega T_1) / \omega T$ at $\omega = k\omega_0$.